Nolpaza

Pantoprazole

NAME OF THE MEDICINAL PRODUCT

Nolpaza gastro-resistant tablets 40 mg

PRODUCT DESCRIPTION AND COMPOSITION

Each gastro-resistant tablet contains 40 mg pantoprazole as pantoprazole sodium sesquihydrate.

The tablets are light yellowish brown, oval, slightly biconvex, film-coated tablets.

Each gastro-resistant tablet contains 36 mg sorbitol (E420).

DOSAGE FORM

Oral Gastro-resistant tablet.

THERAPEUTIC INDICATIONS

- In combination with two appropriate antibiotics (see „Posology") for the eradication of Helicobacter pylori in patients with peptic ulcers with the objective of reducing the recurrence of duodenal and gastric ulcers caused by this microorganism
- Duodenal ulcer
- Gastric ulcer
- Moderate and severe cases of inflammation of the esophagus (reflux esophagitis).
- Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions

POSOLOGY AND METHOD OF ADMINISTRATION

The following information applies unless Nolpaza 40 mg has been otherwise prescribed by your doctor. Please follow these instructions, as otherwise Nolpaza 40 mg may not have the desired effect!

Adults and adolescents 12 years of age and above
<u>Treatment of moderate and severe reflux oesophagitis</u>
One tablet of Nolpaza 40 mg per day. In individual cases the dose may be doubled (increase to 2 tablets Nolpaza 40 mg daily) especially when there has been no response to other treatment.

Adults
<u>Eradication of H. pylori in combination with two appropriate antibiotics</u>
In cases of duodenal or gastric ulcer in which infection with Helicobacter pylori has been confirmed, the microorganism should be eradicated by combination treatment. Depending on the resistance pattern, the following combinations are recommended:

a) Twice daily one tablet Nolpaza 40 mg gastro-resistant tablet
+ twice daily 1000 mg amoxicillin
+ twice daily 500 mg clarithromycin

b) Twice daily one tablet Nolpaza 40 mg gastro-resistant tablet
+ twice daily 500 mg metronidazole
+ twice daily 500 mg clarithromycin

c) Twice daily one tablet Nolpaza 40 mg gastro-resistant tablet
+ twice daily 1000 mg amoxicillin

+ twice daily 500 mg metronidazole

Treatment of gastric and duodenal ulcer
One tablet of Nolpaza 40 mg per day. In individual cases the dose may be doubled (increase to 2 tablets of Nolpaza 40 mg daily) especially when there has been no response to other treatment.

Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions
For the long-term management of Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions patients should start their treatment with a daily dose of 80 mg (2 tablets of Nolpaza 40 mg). Thereafter, the dosage can be titrated up or down as needed using measurements of gastric acid secretion to guide. With doses above 80 mg daily, the dosage should be divided and given twice daily. A temporary increase of the dose above 160 mg pantoprazole is possible but should not be applied longer than required for adequate acid control. Treatment duration in Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs.

Children below 12 years of age
Nolpaza 40 mg is not recommended for use in children below 12 years of age due to limited data in this age group.

Type and duration of treatment
Combination therapy for eradication of Helicobacter pylori infection usually lasts 7 days and can be extended to a maximum of 2 weeks. If after this time further treatment with Nolpaza 40 mg is indicated to ensure that the ulcer heals completely, the dose recommendations for gastric and duodenal ulcers must be observed. In the majority of cases, a duodenal ulcer heals completely within 2 weeks. If a two-week treatment period is not sufficient, healing will be achieved in almost all cases within a further 2 weeks. Gastric ulcers and reflux esophagitis usually require a 4-week course of treatment. If this should be inadequate, healing will in most cases be achieved within a further 4 weeks. Treatment should not exceed 8 weeks as experience with long-term use is limited. Treatment duration in Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs

Instructions for use / handling
Route of administration: oral

Nolpaza 40 mg gastro-resistant tablets must not be chewed or crushed and must be swallowed whole with water. In combination therapy for eradication of Helicobacter pylori infection the second Nolpaza 40 mg tablet should be taken before the evening meal.

Special Patient Populations

Impaired hepatic function
A daily dose of 20 mg pantoprazole (1 tablet of 20 mg pantoprazole or half a vial of 40 mg pantoprazole I.V.) should not be exceeded in patients with severe liver impairment.

Impaired renal function
No dose adjustment is necessary in patients with impaired renal function.

CONTRAINDICATIONS

Nolpaza 40 mg must not be used in combination treatment for eradication Helicobacter pylori in patients with moderate to severe liver or kidney function disturbances since currently no clinical data are available on the efficacy and safety of Nolpaza 40 mg in combination treatment of these patients. Nolpaza 40 mg should generally not be used in cases of known hypersensitivity to the active substance, substituted benzimidazoles, to one of the constituents of Nolpaza 40 mg or to the combination partners.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Hepatic impairment
In patients with severe liver impairment, the liver enzymes should be monitored regularly during treatment with pantoprazole, particularly on long-term use. In the case of a rise of the liver enzymes, the treatment should be discontinued.

Combination therapy
In the case of combination therapy, the summaries of product characteristics of the respective medicinal products should be observed.

Gastric malignancy
Symptomatic response to pantoprazole may mask the symptoms of gastric malignancy and may delay diagnosis. In the presence of any alarm symptom (e. g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded.
Further investigation is to be considered if symptoms persist despite adequate treatment.

Co-administration with HIV protease inhibitors
Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir, due to significant reduction in their bioavailability.

Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should be consider stopping Nolpaza 40 mg. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia
Severe hypomagnesaemia has been reported in patients treated with PPI like Nolpaza 40 mg for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but the may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea
Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be

considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Gastrointestinal infections caused by bacteria
Treatment with Nolpaza 40 mg may lead to a slightly increased risk of gastrointestinal infections caused by bacteria such as Salmonella and Campylobacter or C. difficile.

Vitamin B12 Deficiency
Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with laboratory tests
Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patients(s) are due to have a test on Chromogranin A level, Nolpaza 40 mg treatment should be stopped for at least 5 days before CgA measurements to avoid interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Nolpaza contains 36 mg sorbitol in each tablet.
The additive effect of concomitantly administered products containing sorbitol (or fructose) and dietary intake of sorbitol (or fructose) should be taken into account.
The content of sorbitol in medicinal products for oral use may affect the bioavailability of other medicinal products for oral use administered concomitantly.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Medicinal products with pH-Dependent Absorption Pharmacokinetics
Because of profound and long lasting inhibition of gastric acid secretion, pantoprazole may interfere with the absorption of other medicinal products where gastric pH is an important determinant of oral availability, e.g. some azole antifungals such as ketoconazole, itraconazole, posaconazole and other medicine such as erlotinib.

HIV protease inhibitors
Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir due to significant reduction in their bioavailability.

If the combination of HIV protease inhibitors with a proton pump inhibitor is judged unavoidable, close clinical monitoring (e.g virus load) is recommended. A pantoprazole dose of 20 mg per day should not be exceeded. Dosage of the HIV protease inhibitors may need to be adjusted.

Coumarin anticoagulants (phenprocoumon or warfarin)
Co-administration of pantoprazole with warfarin or phenprocoumon did not affect the pharmacokinetics of warfarin, phenprocoumon or INR. However, there have been reports of increased INR and prothrombin time in patients receiving PPIs and warfarin or phenprocoumon concomitantly. Increases in INR and prothrombin time may lead to abnormal bleeding, and even death. Patients treated with pantoprazole and warfarin or phenprocoumon may need to be monitored for increase in INR and prothrombin time.

Methotrexate

Concomitant use of high dose methotrexate (e.g. 300 mg) and proton-pump inhibitors has been reported to increase methotrexate levels in some patients. Therefore in settings where high-dose methotrexate is used, for example cancer and psoriasis, a temporary withdrawal of pantoprazole may need to be considered.

Other interactions studies
Pantoprazole is extensively metabolized in the liver via the cytochrome P450 enzyme system. The main metabolic pathway is demethylation by CYP2C19 and other metabolic pathways include oxidation by CYP3A4.

Interaction studies with medicinal products also metabolized with these pathways, like carbamazepine, diazepam, glibenclamide, nifedipine, and an oral contraceptive containing levonorgestrel and ethinyl oestradiol, did not reveal clinically significant interactions.

An interaction of pantoprazole with other medicinal products or compounds, which are metabolized using the same enzyme system, cannot be excluded.

Results from a range of interaction studies demonstrate that pantoprazole does not affect the metabolism of active substances metabolised by CYP1A2 (such as caffeine, theophylline), CYP2C9 (such as piroxicam, diclofenac, naproxen), CYP2D6 (such as metoprolol), CYP2E1 (such as ethanol), or does not interfere with p-glycoprotein related absorption of digoxin.

There were no interactions with concomitantly administered antacids.

Interaction studies have also been performed by concomitantly administering pantoprazole with the respective antibiotics (clarithromycin, metronidazole, amoxicillin). No clinically relevant interactions were found.

Medicinal products that inhibit or induce CYP2C19
Inhibitors of CYP2C19 such as fluvoxamine could increase the systemic exposure of pantoprazole. A dose reduction may be considered for patients treated long-term with high doses of pantoprazole, or those with hepatic impairment.
Enzyme inducers affecting CYP2C19 and CYP3A4 such as rifampicin and St John´s wort (*Hypericum perforatum*) may reduce the plasma concentrations of PPIs that are metabolized through these enzyme systems.

PREGNANCY AND LACTATION

Pregnancy
A moderate amount of data on pregnant women (between 300-1000 pregnancy outcomes) indicate no malformative or feto/ neonatal toxicity of Nolpaza 40 mg. Animal studies have shown reproductive toxicity. As a precautionary measure, it is preferable to avoid the use of Nolpaza 40 mg during pregnancy.

Breast-feeding
Animal studies have shown excretion of pantoprazole in breast milk. There is insufficient information on the excretion of pantoprazole in human milk but excretion into human milk has been reported. A risk to the newborns/infants cannot be excluded. Therefore, a decision on whether to discontinue breast-feeding or to discontinue/abstain from Nolpaza 40 mg therapy taking into account the benefit of breast-feeding for the child, and the benefit of Nolpaza 40 mg therapy for the woman.

Fertility
There was no evidence of impaired fertility following the administration of pantoprazole in animal studies

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Pantoprazole has no or negligible influence on the ability to drive and use machines.
Adverse drug reactions, such as dizziness and visual disturbances may occur. If affected, patients should not drive or operate machines.

UNDESIRABLE EFFECTS

Approximately 5% of patients can be expected to experience adverse drug reactions (ADRs). The most commonly reported ADRs are diarrhoea and headache, both occurring in approximately 1% of patients.

For all adverse reactions reported from post-marketing experience, it is not possible to apply any adverse reaction frequency and therefore they are mentioned with a "not known" frequency.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1. Adverse reactions with pantoprazole

Frequency / System Organ Class	Common	Uncommon	Rare	Very rare	Not known
Blood and lymphatic system disorders			Agranulocytosis	thrombocytopenia; leukopenia; pancytopenia	
Immune system disorders			hypersensitivity (including anaphylactic reactions and anaphylactic shock)		
Metabolism and nutrition disorders			hyperlipidaemias and lipid increases (triglycerides, cholesterol); weight changes		hyponatraemia; hypomagnesaemia; vitamin B12 deficiency; hypocalcaemia [(1)]; hypokalaemia
Psychiatric disorders		sleep disorders	depression (and all aggravations)	disorientation (and all aggravations)	hallucination; confusion (especially in pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence)
Nervous system disorders		headache; dizziness	taste disorders		Parasthesia
Eye disorders			disturbances in vision / blurred vision		
Gastrointestinal disorders	Fundic gland polyps (benign)	diarrhoea; nausea/vomiting; abdominal distension and			Microscopic colitis

		bloating; constipation; dry mouth; abdominal pain and discomfort			
Hepatobiliary disorders		liver enzymes increased (transaminases, γ-GT)	bilirubin increased		hepatocellular injury; jaundice; hepatocellular failure
Skin and subcutaneous tissue disorders		rash/exanthema/ eruption; pruritus	urticaria; angioedema		Stevens-Johnson syndrome; Lyell's syndrome; erythema multiforme; photosensitivity Subacute cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders		fracture of the hip, wrist or spine	arthralgia; myalgia	myalgia	muscle spasm (2)
Renal and urinary disorders					interstitial nephritis (with possible progression to renal failure)
Reproductive system and breast disorders			gynaecomastia		
General disorders and administration site conditions		asthenia; fatigue and malaise	body temperature increased; oedema peripheral		
Infections and infestations					*Clostridium difficile* associated diarrhea

1. Hypocalcemia in association with hypomagnesemia
2. Muscle spasm as a consequence of electrolyte disturbance

OVERDOSE

There are no known symptoms of overdose in man.
Systemic exposure with up to 240 mg administered intravenously over 2 minutes, were well tolerated.
As pantoprazole is extensively protein bound, it is not readily dialysable.

PHARMACOLOGICAL PARTICULARS

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: proton pump inhibitors; ATC code: A02BC02.

Mechanism of action
Pantoprazole is a substituted benzimidazole which inhibits the secretion of hydrochloric acid in the stomach by specific blockade of the proton pumps of the parietal cells.

Pantoprazole is converted to its active form in the acidic environment in the parietal cells where it inhibits the H^+, K^+-ATPase enzyme, i.e. the final stage in the production of hydrochloric acid in the stomach. The inhibition is dose-dependent and affects both basal and stimulated acid secretion. In most patients, freedom from symptoms is achieved within 2 weeks. As with other proton pump inhibitors and H_2 receptor inhibitors, treatment with pantoprazole reduces acidity in the stomach and thereby increases gastrin in proportion to the reduction in acidity. The increase in gastrin is reversible. Since pantoprazole binds to the enzyme distal to the cell receptor level, it can inhibit hydrochloric acid secretion independently of stimulation by other substances (acetylcholine, histamine, gastrin). The effect is the same whether the product is given orally or intravenously.

The fasting gastrin values increase under pantoprazole. On short-term use, in most cases they do not exceed the upper limit of normal. During long-term treatment, gastrin levels double in most cases. An excessive increase, however, occurs only in isolated cases. As a result, a mild to moderate increase in the number of specific endocrine (ECL) cells in the stomach is observed in a minority of cases during long-term treatment (simple to adenomatoid hyperplasia). However, according to the studies conducted so far, the formation of carcinoid precursors (atypical hyperplasia) or gastric carcinoids as were found in animal experiments have not been observed in humans.

An influence of long-term (exceeding one year) treatment with pantoprazole on endocrine parameters of the thyroid cannot be completely ruled out according to results in animal studies.

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.
Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

PHARMACOKINETIC PROPERTIES

Absorption
Pantoprazole is rapidly absorbed and the maximal plasma concentration is achieved even after one single 40 mg oral dose. On average at about 2.5 h p.a. the maximum serum concentrations of about 2 - 3 µg/ml are achieved, and these values remain constant after multiple administration.

Pharmacokinetics does not vary after single or repeated administration. In the dose range of 10 to 80 mg, the plasma kinetics of pantoprazole are linear after both oral and intravenous administration.

The absolute bioavailability from the tablet was found to be about 77 %. Concomitant intake of food had no influence on AUC, maximum serum concentration and thus bioavailability. Only the variability of the lag-time will be increased by concomitant food intake.

Distribution
Pantoprazole's serum protein binding is about 98 %. Volume of distribution is about 0.15 l/kg.

Biotransformation
The substance is almost exclusively metabolized in the liver. The main metabolic pathway is demethylation by CYP2C19 with subsequent sulphate conjugation; other metabolic pathway includes oxidation by CYP3A4.

Elimination
Terminal half-life is about 1 hour and clearance is about 0.1 l/h/kg. There were a few cases of subjects with delayed elimination. Because of the specific binding of pantoprazole to the proton pumps of the parietal cell the elimination half-life does not correlate with the much longer duration of action (inhibition of acid secretion).
Renal elimination represents the major route of excretion (about 80 %) for the metabolites of pantoprazole; the rest is excreted with the faeces. The main metabolite in both the serum and urine is

desmethylpantoprazole which is conjugated with sulphate. The half-life of the main metabolite (about 1.5 hours) is not much longer than that of pantoprazole.

Special populations

Poor metabolisers
Approximately 3 % of the European population lack a functional CYP2C19 enzyme and are called poor metabolisers. In these individuals the metabolism of pantoprazole is probably mainly catalysed by CYP3A4. After a single-dose administration of 40 mg pantoprazole, the mean area under the plasma concentration-time curve was approximately 6 times higher in poor metabolisers than in subjects having a functional CYP2C19 enzyme (extensive metabolisers). Mean peak plasma concentrations were increased by about 60 %. These findings have no implications for the posology of pantoprazole.

Renal impairment
No dose reduction is recommended when pantoprazole is administered to patients with impaired renal function (including dialysis patients). As with healthy subjects, pantoprazole's half-life is short. Only very small amounts of pantoprazole are dialyzed. Although the main metabolite has a moderately delayed half-life (2 - 3 h), excretion is still rapid and thus accumulation does not occur.

Hepatic impairment
Although for patients with liver cirrhosis (classes A and B according to Child) the half-life values increased to between 7 and 9 h and the AUC values increased by a factor of 5 - 7, the maximum serum concentration only increased slightly by a factor of 1.5 compared with healthy subjects.

Older people
A slight increase in AUC and C_{max} in elderly volunteers compared with younger counterparts is also not clinically relevant.

Paediatric population
Following administration of single oral doses of 20 or 40 mg pantoprazole to children aged 5 - 16 years AUC and C_{max} were in the range of corresponding values in adults.

Following administration of single i.v. doses of 0.8 or 1.6 mg/kg pantoprazole to children aged 2 - 16 years there was no significant association between pantoprazole clearance and age or weight. AUC and volume of distribution were in accordance with data from adults.

LIST OF EXCIPIENTS

Tablet core
mannitol
crospovidone
sodium carbonate
sorbitol
calcium stearate

Film coating
hypromellose
povidone
titanium dioxide (E 171)
yellow iron oxide (E 172)
propylene glycol
methacrylic acid - ethyl acrylate copolymer (1:1) dispersion 30 per cent
macrogol 6000
talc

SPECIAL PRECAUTIONS FOR STORAGE

Do not store above 30 °C. Store in the original package in order to protect from moisture.

PRESENTATION

Blister pack 28 gastro-resistant tablets, in a box.

PRODUCT REGISTRATION HOLDER

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